

Overview of Explainable artificial intelligence

Subject: Explainable artificial intelligence

1.

Background Machine learning (ML) algorithms used in AI can be categorized as white-box or black-box. White-box models provide results that are understandable to experts in the domain. Black-box models, on the other hand, are extremely hard to explain and may not be understood even by domain experts. XAI algorithms follow the three principles of transparency, interpretability, and explainability.

2.

Technical complexity A fundamental barrier to making AI systems explainable is the technical complexity of such systems. End users often lack the coding knowledge required to understand software of any kind. Current methods used to explain AI are mainly technical ones, geared toward machine learning engineers for debugging purposes, rather than toward the end users who are ultimately affected by the system, causing "a gap between explainability in practice and the goal of transparency". Proposed solutions to address the issue of technical complexity include either promoting the coding education of the general public so technical explanations would be more accessible to end users, or providing explanations in layperson terms. The solution must avoid oversimplification. It is important to strike a balance between accuracy – how faithfully the explanation reflects the process of the AI system – and explainability – how well end users understand the process. This is a difficult balance to strike, since the complexity of machine learning makes it difficult for even ML engineers to fully understand, let alone non-experts.

3.

Participatory budgeting Yang, Hausladen, Peters, Pournaras, Fricker and Helbing present an empirical study of explainability in participatory budgeting. They compared the greedy and the equal shares rules, and three types of explanations: mechanism explanation (a general explanation of how the aggregation rule works given the voting input), individual explanation (explaining how many voters had at least one approved project, at least 10000 CHF in approved projects), and group explanation (explaining how the budget is distributed among the districts and topics). They compared the perceived trustworthiness and fairness of greedy and equal shares, before and after the explanations. They found out that, for MES, mechanism explanation yields the highest increase in perceived fairness and trustworthiness; the second-highest was Group explanation. For Greedy, Mechanism explanation increases perceived trustworthiness but not fairness, whereas Individual explanation increases both perceived fairness and trustworthiness. Group explanation decreases the perceived fairness and trustworthiness.

4.

Adversarial parties By making an AI system more explainable, we also reveal more of its inner workings. For example, the explainability method of feature importance identifies features or variables that are most important in determining the model's output, while the influential samples method identifies the training samples that are most influential in determining the output, given a particular

input. Adversarial parties could take advantage of this knowledge. For example, competitor firms could replicate aspects of the original AI system in their own product, thus reducing competitive advantage. An explainable AI system is also susceptible to being "gamed"—influenced in a way that undermines its intended purpose. One study gives the example of a predictive policing system; in this case, those who could potentially "game" the system are the criminals subject to the system's decisions. In this study, developers of the system discussed the issue of criminal gangs looking to illegally obtain passports, and they expressed concerns that, if given an idea of what factors might trigger an alert in the passport application process, those gangs would be able to "send guinea pigs" to test those triggers, eventually finding a loophole that would allow them to "reliably get passports from under the noses of the authorities".

5.

Explainability in social choice Explainability was studied also in social choice theory. Social choice theory aims at finding solutions to social decision problems, that are based on well-established axioms. Ariel D. Procaccia explains that these axioms can be used to construct convincing explanations to the solutions. This principle has been used to construct explanations in various subfields of social choice.

6.

Knowledge localization Large language models (LLMs), such as transformer-based models (GPT), appear to use general knowledge when generating output (completing inference). This capability raises the question of how, exactly, such knowledge is stored within the model. Research has suggested that the model's MLP component (the feed-forward layers) is the main site in which knowledge is stored, encoding information through associative links that function as key–value memories: each key corresponds to textual patterns in the training data, while each value induces a distribution over the output vocabulary. A 2022 study, aimed at locating where knowledge resides in the model, employed a technique known as Causal Tracing. In tasks requiring general knowledge, the researchers injected noise into the hidden activations of the model, preventing it from completing the task. They then restored the clean activation values (taken from a noise-free run) to a different part of the model each time, observing when the model regained its ability to produce the correct answer. Based on these results, the authors concluded that factual knowledge is stored primarily in the MLP components of the model's middle layers. They further proposed that model editing would be most effective in those regions, though this claim was later called into question. Later studies suggest that, in most cases, factual information is distributed across the model rather than localized within a single layer. According to one version of this view, different layers encode different aspects of the same association. For example, a question about the capital of Japan may activate representations related to "Japan" in one layer and representations corresponding to "capital cities" in another; the combination of these representations yields the concept of "Tokyo".

7.

By just tracing through the dependency structure the problem solver can construct the following explanation: "Socrates died because he was mortal and drank poison, and all mortals die when they drink poison. Socrates was mortal because he was a man and all men are mortal. Socrates drank poison because he held dissident beliefs, the government was conservative, and those holding conservative dissident beliefs under conservative governments must drink poison." By the 1990s researchers began studying whether it is possible to meaningfully extract the non-hand-coded rules being generated by

opaque trained neural networks. Researchers in clinical expert systems who created neural network-powered decision support for clinicians sought to develop dynamic explanations that allow these technologies to be more trusted and trustworthy in practice. In the 2010s public concerns about racial and other bias in the use of AI for criminal sentencing decisions and findings of creditworthiness may have led to increased demand for transparent artificial intelligence. As a result, many academics and organizations are developing tools to help detect bias in their systems. Marvin Minsky et al. raised the issue that AI can function as a form of surveillance, with the biases inherent in surveillance, suggesting HI (Humanistic Intelligence) as a way to create a more fair and balanced "human-in-the-loop" AI. Explainable AI has been recently a new topic researched amongst the context of modern deep learning. Modern complex AI techniques, such as deep learning, are naturally opaque. To address this issue, methods have been developed to make new models more explainable and interpretable. This includes layerwise relevance propagation (LRP), a technique for determining which features in a particular input vector contribute most strongly to a neural network's output, although this technique has been shown to suffer from several important issues. Other techniques explain some particular prediction made by a (nonlinear) black-box model, a goal referred to as "local interpretability". We still today cannot explain the output of today's DNNs without the new explanatory mechanisms, we also can't by the neural network, or external explanatory components. There is also research on whether the concepts of local interpretability can be applied to a remote context, where a model is operated by a third-party. There has been work on making glass-box models which are more transparent to inspection. This includes decision trees, Bayesian networks, sparse linear models, and more. The Association for Computing Machinery Conference on Fairness, Accountability, and Transparency (ACM FAccT) was established in 2018 to study transparency and explainability in the context of socio-technical systems, many of which include artificial intelligence. Some techniques allow visualisations of the inputs to which individual software neurons respond to most strongly. Several groups found that neurons can be aggregated into circuits that perform human-comprehensible functions, some of which reliably arise across different networks trained independently. There are various techniques to extract compressed representations of the features of given inputs, which can then be analysed by standard clustering techniques. Alternatively, networks can be trained to output linguistic explanations of their behaviour, which are then directly human-interpretable. Model behaviour can also be explained with reference to training data—for example, by evaluating which training inputs influenced a given behaviour the most, or by approximating its predictions using the most similar instances from the training data. The use of explainable artificial intelligence (XAI) in pain research, specifically in understanding the role of electrodermal activity for automated pain recognition: hand-crafted features and deep learning models in pain recognition, highlighting the insights that simple hand-crafted features can yield comparative performances to deep learning models and that both traditional feature engineering and deep feature learning approaches rely on simple characteristics of the input time-series data.